University of Prishtina “Hasan Prishtina”

Faculty of Electrical and Computer Engineering

Second Intrusion Report

Team 18: CyberBufi

Prishtinë, 2026

Table of Contents

Table of Figures 3
1. Flag 2 - Linux Root-Level Post-Exploitation Activity 4
1.1. Flags Identified 4
1.2. Active Systems and Services 4
1.3. Evidence Collected 4
2. Flag 3 - SSH Manipulation and Persistence Attempts 5
2.1. Flags Identified 5
2.2. Active Systems and Services 6
2.3. Evidence Collected 6
3. Flag 4 - Linux User Account Manipulation and Evidence Tampering 7
3.1. Flags Identified 7
3.2. Active Systems and Services 7
3.3. Evidence Collected 7

Table of Figures

Figure 1: Root terminal activity showing sensitive directory access and flag file removal, indicating potential post-exploitation behavior. 5
Figure 2: Root terminal session showing SSH configuration inspection, login record analysis, firewall rule changes, and artifact manipulation indicating potential post-exploitation activity on a Linux system. 7
Figure 3: Root terminal session displaying login record analysis, artifact manipulation, and user account management actions indicating potential post exploitation activity on a Linux system... 8

1. Flag 1 - Linux Root-Level Post-Exploitation Activity

1.1. Flags Identified

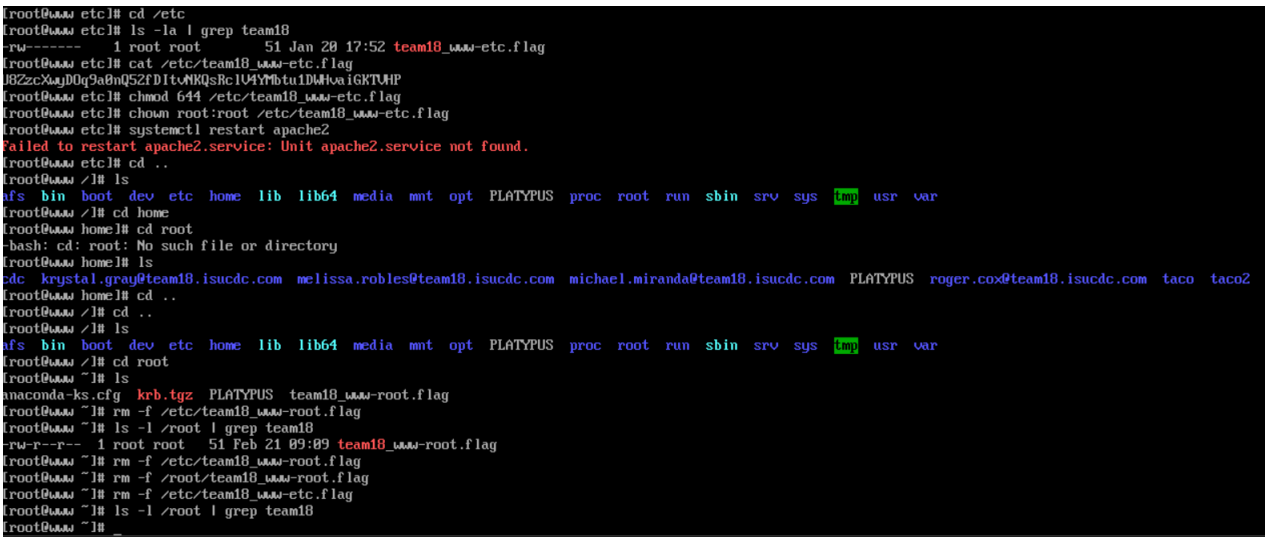
The session reveals multiple indicators of suspicious behavior, including privileged access to sensitive directories such as **/etc** and **/root**, enumeration of system files, modification of file permissions and ownership, and deletion of flag artifacts. These actions suggest post-exploitation activity, reconnaissance within critical system areas, and attempts to remove evidence in order to evade detection.

1.2. Active Systems and Services

The system appears to be a Linux host operating with **root-level privileges**, exposing critical filesystem locations and administrative capabilities. The presence of service management commands indicates interaction with system services, while directory exploration demonstrates access to core operating system components and user-related resources that could facilitate persistence or lateral movement.

1.3. Evidence Collected

Evidence from the terminal output includes directory listings, discovery of flag files within protected locations, execution of permission and ownership modification commands, and attempts to remove files from sensitive directories. These actions collectively demonstrate unauthorized system interaction and potential efforts to manipulate or conceal artifacts within the host environment.



```
[root@www etc]# cd /etc
[root@www etc]# ls -la | grep team18
-rw-------    1 root root       51 Jan 20 17:52 team18_www-etc.flag
[root@www etc]# cat /etc/team18_www-etc.flag
J8ZzcXwyDOq9a0nQ52fDItvMKQsRclV4YMbtu1DWHvaiGKTVHP
[root@www etc]# chmod 644 /etc/team18_www-etc.flag
[root@www etc]# chown root:root /etc/team18_www-etc.flag
[root@www etc]# systemctl restart apache2
Failed to restart apache2.service: Unit apache2.service not found.
[root@www etc]# cd ..
[root@www /]# ls
afs  bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  PLATYPUS  proc  root  run  sbin  srv  sys  tmp  usr  var
[root@www /]# cd home
[root@www home]# cd root
-bash: cd: root: No such file or directory
[root@www home]# ls
cdc  krystal.gray@team18.isucdc.com  melissa.robles@team18.isucdc.com  michael.miranda@team18.isucdc.com  PLATYPUS  roger.cox@team18.isucdc.com  taco  taco2
[root@www home]# cd ..
[root@www /]# cd ..
[root@www /]# ls
afs  bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  PLATYPUS  proc  root  run  sbin  srv  sys  tmp  usr  var
[root@www /]# cd root
[root@www ~]# ls
anaconda-ks.cfg  krb.tgz  PLATYPUS  team18_www-root.flag
[root@www ~]# rm -f /etc/team18_www-root.flag
[root@www ~]# ls -l /root | grep team18
-rw-r--r--  1 root root   51 Feb 21 09:09 team18_www-root.flag
[root@www ~]# rm -f /etc/team18_www-root.flag
[root@www ~]# rm -f /root/team18_www-root.flag
[root@www ~]# rm -f /etc/team18_www-etc.flag
[root@www ~]# ls -l /root | grep team18
[root@www ~]# _
```

Figure 1: Root terminal activity showing sensitive directory access and flag file removal, indicating potential post-exploitation behavior.

2. Flag 2 - SSH Manipulation and Persistence Attempts

2.1. Flags Identified

Terminal activity reveals several indicators of compromise, including inspection and modification of critical **SSH configuration files**, enabling root login and password-based authentication, and attempts to remove configuration artifacts. Additional suspicious actions include reviewing shell history, creating and deleting flag files within privileged directories, examining login records, and applying firewall rules to block external IP addresses. These behaviors indicate post-exploitation activity and potential attempts to establish persistence or control system access.

2.2. Active Systems and Services

The system appears to be a Linux host operating with **root privileges** and active **SSH services** configured for remote access. Authentication services, firewall controls implemented through **iptables**, and system logging mechanisms were operational, indicating a fully functional environment in which administrative commands and service configurations could be modified.

2.3. Evidence Collected

Collected evidence includes the discovery of sensitive SSH configuration files, shell history records indicating executed commands, login session logs showing external access attempts, and firewall rule modifications targeting a specific remote IP address. The creation and removal of artifacts within protected directories further demonstrate unauthorized interaction and possible attempts to conceal malicious activity.

```
[root@www ~]# /etc/
-bash: /etc/: Is a directory
[root@www ~]# sudo find /etc -type f -mmin -120
/etc/pam.d/common-auth
/etc/shadow
/etc/ssh/sshd_config.d/custom.conf
/etc/ssh/sshd_config
[root@www ~]# cat /etc/ssh/sshd_config.d/custom.conf
PermitRootLogin yes
PasswordAuthentication yes
PubkeyAuthentication yes
[root@www ~]# sudo rm -f /etc/ssh/sshd_config.d/custom.conf
rm: cannot remove '/etc/ssh/sshd_config.d/custom.conf': Operation not permitted
[root@www ~]# cat /root/.bash_history
cd
ls
echo "hDhOVX92fYsVJu9M8gQPBoGwdJ3DztDJSyr314YWnhueI8nMGD" > team18_www-root.flag
exit
[root@www ~]# rm -f /root/team18_www-root.flag
[root@www ~]# last -a | head -n 10
roger.co pts/0        Sat Feb 21 09:08 - 09:19  (00:10)     68.32.238.130
cdc      tty1         Tue Feb 17 18:32   still logged in
reboot   system boot  Tue Jan 20 17:56   still running      5.14.0-611.20.1.el9_7.x86_64

wtmp begins Tue Jan 20 17:56:15 2026
[root@www ~]# iptables -A INPUT -S 68.32.238.130 -J DROP
iptables v1.8.10 (nf_tables): Cannot use -S with -A
Try `iptables -h' or 'iptables --help' for more information.
[root@www ~]# iptables -A INPUT -s 68.32.238.130 -j DROP
[root@www ~]# userdel -r roger.co
userdel: user 'roger.co' does not exist
[root@www ~]#
```

Figure 2: Root terminal session showing SSH configuration inspection, login record analysis, firewall rule changes, and artifact manipulation indicating potential post-exploitation activity on a Linux system.

3. Flag 3 - Linux User Account Manipulation and Evidence Tampering

3.1. Flags Identified

The terminal session shows indicators of suspicious privileged activity, including reading flag files from protected directories, creation of new flag artifacts, review of login records revealing external IP sessions, and attempts to delete user accounts. These behaviors suggest post exploitation actions, reconnaissance of system activity, and potential attempts to manipulate or conceal evidence.

3.2. Active Systems and Services

The system appears to be a Linux host operating with root level privileges, with authentication services, system logging mechanisms, and user management services active. The presence of login tracking and account management commands indicates interaction with core identity and access services within the environment.

3.3. Evidence Collected

The investigation indicates unauthorized root level interaction involving access to sensitive directories, artifact creation and deletion, and analysis of login history showing remote sessions from external IP addresses. Attempts to remove user accounts and manipulate system artifacts further suggest post exploitation behavior and possible

efforts to evade detection. Overall, the activity reflects a compromised host scenario involving reconnaissance, persistence attempts, and evidence manipulation.

```
mysql
cat /etc/*flag
echo 6QTfT46JGBF20AedG322Zwm1xIDAsVRRpKSleVtsiMaWA0Kv00 > team18_mgmt-root.flag
cat team18_mgmt-root.flag
exit
[root@mgmt ~]# last -a | head -n 10
roger.co pts/0        Sat Feb 21 08:03 - 08:09  (00:05)     68.32.238.130
cdc      tty1         Fri Feb 20 13:45   still logged in
reboot   system boot  Fri Feb 20 13:45   still running      4.18.0-305.el8.x86_64
reboot   system boot  Fri Feb 20 13:40   still running      4.18.0-305.el8.x86_64
cdc      tty1         Fri Feb 20 13:28 - 13:39  (00:11)
reboot   system boot  Fri Feb 20 13:26 - 13:40  (00:13)     4.18.0-305.el8.x86_64
cdc      tty1         Fri Feb 20 13:24 - 13:25  (00:00)
reboot   system boot  Fri Feb 20 13:24 - 13:25  (00:01)     4.18.0-305.el8.x86_64
roger.co pts/0        Thu Feb 19 09:30 - 09:30  (00:00)     12.110.205.163
melissa. pts/0        Thu Feb 19 09:25 - 09:25  (00:00)     68.32.254.50
[root@mgmt ~]# userdel -r roger.co
userdel: user 'roger.co' does not exist
[root@mgmt ~]# userdel -r melissa.
userdel: user 'melissa.' does not exist
```

Figure 3: Root terminal session displaying login record analysis, artifact manipulation, and user account management actions indicating potential post exploitation activity on a Linux system.